



Society of Petroleum Engineers

SPE-229887-MS

Enhancing Coiled Tubing Water Shut-Off Operations: Lessons Learned and Best Practices for Zonal Isolation

A. Alaedan, H. Alturaiki, A. Almazrou, K. Redouane, and A. AlAli, Wassim Kharrat TAQA Well Solutions

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229887-MS](https://doi.org/10.2118/229887-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Coiled tubing (CT) water shut off treatments have been proven as an effective method in reducing water cut in production wells. This paper introduces innovative design and execution techniques for CT water shut off treatments. It also highlights the importance of candidate and treatment selection for different completions to ensure successful water shut-off operations. Additionally, the paper addresses the ways to optimize water shut-off operations and mitigate all the associated risks.

The first method addresses wells completed with Inflow Control Devices (ICDs), emphasizing the necessary precautions to prevent unintended production loss in non-target zones during treatment. The second method focuses on wells with perforated completions and compares its complexity and efficiency with ICD-based designs. Additional completion types—such as screens and open-hole configurations—are also discussed, highlighting the fact that the basic treatment principles remain the same, while the execution complexity varies significantly. The paper also outlines optimal operating procedures and demonstrates how real-time downhole monitoring systems—integrated through both CT and Electrical-Wireline (E-line)—enhance treatment design, volume accuracy, execution control, and post-treatment evaluation.

The paper provides insight into how the integration of mechanical and chemical isolation methods with real-time downhole monitoring systems can provide the best solution for operational execution of water shut off in any type of completion.

Several observations were made such as the implementation of a pre-treatment logging survey to accurately identify water-producing zones, which can ultimately optimize the treatment volume. Additionally, it is crucial to ensure proper isolation of completion components or downhole mechanical isolation tools to prevent water shut-off treatments from invading other hydrocarbon producing zones. Moreover, the use of an on-location laboratory was critical for optimizing the chemical recipe based on real-time downhole acquired data.

Cleanout operations were found to be essential to guarantee the correct setting of mechanical isolation downhole tools and to provide a clear path for fishing tools to retrieve the mechanical isolation tool. The treatment deployment process was concluded using an appropriate bottom hole assembly (BHA), taking into consideration whether upper zone isolation was necessary or not. The downhole monitoring system proved

valuable in enabling real-time depth correlations, as well as monitoring downhole pressure and temperature during treatment placement, which were vital for the successful execution of these operations.

This paper represents CT innovative design and execution techniques for water zonal isolation in oil and gas wells that can be utilized for maintaining high production levels while reducing the operational cost of these wells. Moreover, it provides a clear path to rig-less water shut-off solutions.

Introduction

Water is a vital resource for life on Earth, but in the context of oil and gas production, excessive water production poses a significant operational and economic challenge. As hydrocarbon reservoirs mature and production continues, water cut often increases—leading to higher costs for separation, treatment, and disposal, and reduced return on investment. One of the common methods to sustain reservoir pressure and improve recovery is waterflooding; however, it can sometimes lead to unintended water breakthrough for some wells. Additionally, wellbore integrity issues such as casing leaks or completion failures can introduce water from non-target zones, further increasing water production. Surface management of produced water is expensive, making it more effective and economical to address the problem at its source—downhole.

Water management downhole can be approached through several solutions, including ICDs, cement squeezes when the source is well-identified, sidetracking in severe cases, or full plug and abandonment when no viable remediation is possible. Another increasingly adopted approach involves non-polymer-based treatments to isolate water-producing zones. While effective, such treatments can be expensive and technically challenging without rig-less access. Coiled tubing provides a cost-efficient and precise rig-less method, allowing operators to reach the problematic zone and place the treatment with high accuracy. Recent advances in non-polymer chemistry have led to the development of temperature-activated gels ([Almohsin et al, 2025](#)), making bottomhole temperature a critical parameter for treatment success. Deploying a well-designed BHA with real-time telemetry is essential to ensure accurate placement and activation of the treatment fluid.

This paper outlines best practices for conducting coiled tubing-based water shut-off operations using water shut-off treatments across various completion types, including wells with ICDs, perforated intervals, and screens. It emphasizes the critical role of BHA design with real-time telemetry for treatment accuracy, as well as strategies to optimize placement, reduce treatment volume, and improve long-term zonal isolation. By consolidating operational lessons and technical considerations, this paper serves as a practical guide for engineers and planners seeking efficient, rigless water control solutions in mature fields.

Field Background and Problem Statement

Water shut-off campaigns were conducted across a wide range of wells, each with unique completion architectures and operational constraints especially minimum restrictions of the completion jewelry as it limits the type of BHA that can be utilized. These variations significantly impact the approach, risk profile, and effectiveness of downhole treatment operations. A major challenge across all completion types is achieving selective isolation of water-producing intervals while preserving hydrocarbon production from adjacent zones. This section outlines the most common completion types encountered and the associated challenges faced during rig-less coiled tubing shut-off operations.

Completion Types Encountered Starting from the simplest

Perforated Cased Hole Completions. Perforated cased hole wells as show in [figure 1](#) offer the most straightforward environment for water shut-off operations, primarily due to the ability to selectively access and treat specific zones—especially when supported by accurate production logs (PLT) data. However, incorrect identification of the water-producing intervals can result in ineffective treatments or unintended damage to hydrocarbon-producing zones. The complexity increases significantly in wells with bad cement

and/or hydraulic or natural fractures, as the high conductivity of these features can cause inadequate fluid placement and premature loss of the treatment fluid into the formation, preventing effective zone isolation. In such cases, a tailored blend of polymeric and non-polymeric shut-off materials may be necessary to achieve complete sealing.

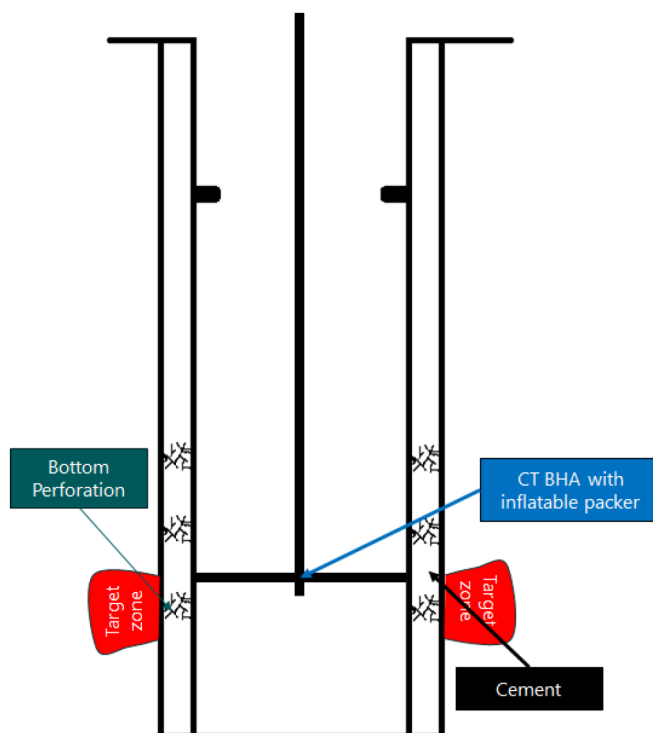


Figure 1—Operational Configuration for Water Shut-Off in a Perforated Liner Completion

Inflow Control Devices (ICDs) or Frac Ports. ICDs and frac ports are widely used in multistage or compartmentalized completions to regulate inflow from various reservoir zones as shown in [figure 2](#). Passive ICDs use fixed nozzles or flow channels to restrict high-rate zones and, in some designs, provide greater resistance to water than oil—helping delay water breakthrough without requiring mechanical operation. However, these devices are not adjustable after installation, limiting the ability to intervene directly in specific intervals. Some completions incorporate sleeves or ports that can be mechanically shifted using through-tubing tools, though they still lack real-time control. These limitations make selective water shut-off particularly challenging, requiring highly accurate treatment placement. Ensuring the presence and effectiveness of open hole packers or other interzonal barriers is critical to confine the treatment to the water-producing zone. Similarly to perforated wells, the presence of natural or induced fractures can lead to uncontrolled fluid migration, often necessitating a combination of polymeric and non-polymeric materials to achieve effective isolation.

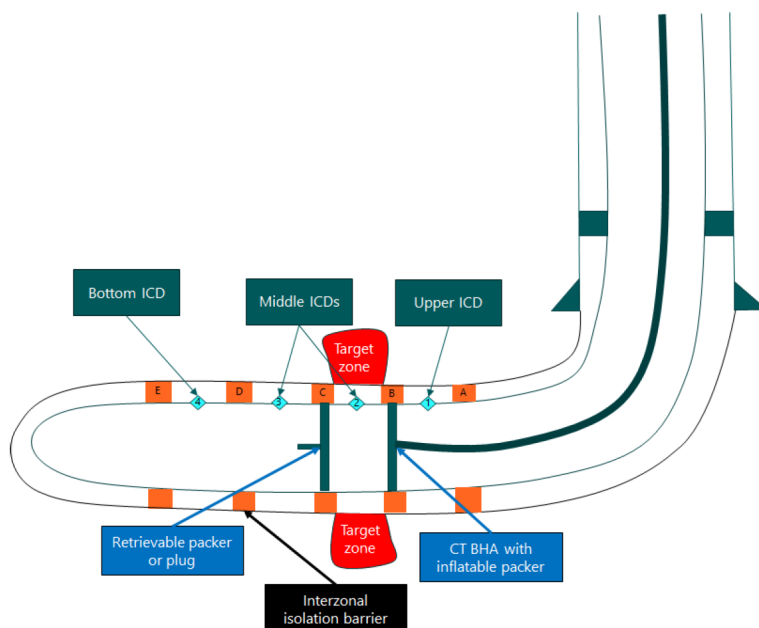


Figure 2—Operational Configuration for Water Shut-Off in a Multistage or ICD Completion

Screen-Only Completions. Commonly used in unconsolidated formations, screen-only completions—whether standalone or gravel-packed—pose significant challenges for water shut-off operations. Mechanical isolation is difficult due to the continuous nature of the screen, which limits access to individual zones. Additionally, production logging in these wells often lacks the resolution needed for confident zonal diagnosis, complicating treatment design. Effective water shut-off in these completions is generally only feasible when multiple screen sections are installed with reliable interzonal barriers, allowing selective treatment while maintaining hydrocarbon production from unaffected zones.

Openhole (Barefoot) Completions. Openhole completions present the greatest challenge for water shut-off operations due to the absence of casing or liners across the reservoir section. This lack of confinement makes the zonal isolation inherently difficult. Mechanical tools have limited anchoring options and success rate, and without natural barriers—such as low-permeability streaks—chemical treatments are prone to uncontrolled placement and crossflow. Effective diversion strategies become essential to confine the treatment to the target zone. Another common issue is borehole washouts, which can prevent proper setting of isolating BHAs. For this reason, a detailed caliper log is critical to identify suitable intervals where the bottomhole assembly can be reliably seated.

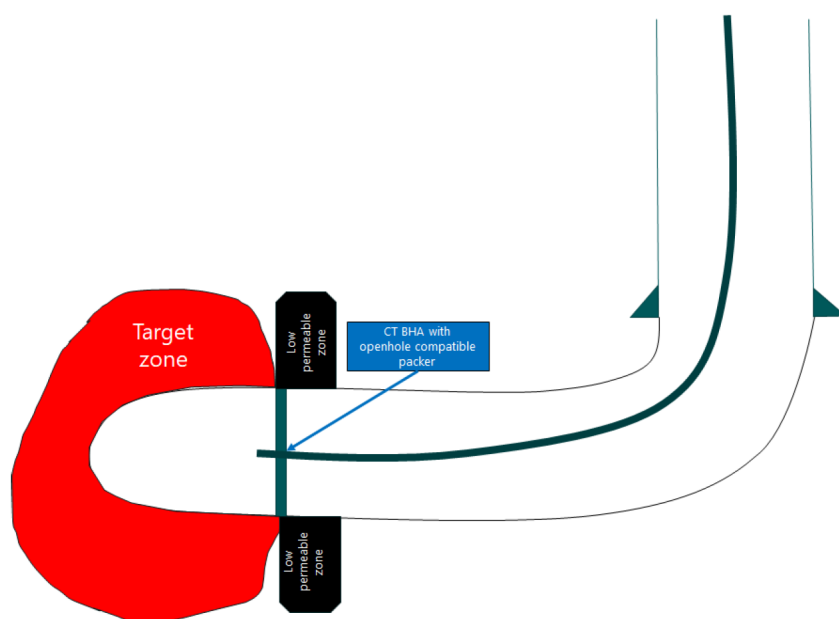


Figure 3—Operational Configuration for Water Shut-Off in an Openhole Completion

Operational Challenges

Water shut-off operations using rig-less coiled tubing interventions face a wide range of challenges, many of which are dictated by the complexity and limitations of the completion architecture. Across various campaigns, the following recurring operational issues were observed:

- **Proper segmentation of Water-Producing Zones:** In completions lacking short zonal segmentation—such as screens—diagnosing and isolating the source of water production remains a key challenge. extended zonal isolation can lead to ineffective treatments and potential loss to hydrocarbon-producing intervals.
- **Limited Mechanical Isolation Options:** Open hole completions is a big challenge for mechanical isolation tools. In fact, This type of completion increases the risk of poor tool setting, especially in the presence of washouts or irregular borehole geometry.
- **Crossflow and Fluid Channeling Risks:** In the absence of effective barriers between reservoir zones, treatment fluids may migrate uncontrollably, especially in fractured formations or open hole environments. This undermines treatment effectiveness and complicates post-job evaluation.
- **Precision in Treatment Volume and Placement:** The success of water shut-off treatments relies heavily on the accurate calculation of treatment volume and controlled fluid placement. Over-displacement can lead to loss of shut-off material into hydrocarbon producing zones, while under-displacement may leave water-producing interval untreated.
- **Completion Restrictions Limiting BHA Design:** Narrow internal diameters, internal profile restrictions, and minimum jewelry clearances within the completion string constrain the size and type of BHA that can be deployed. This often requires customized BHA designs to accommodate both mechanical and chemical treatment needs.
- **Influence of Target Zone Location on BHA Selection:** The location of the water-producing zone significantly impacts the type of isolation strategy used. In fact, when the target zone is located at the very bottom or at the very top of the well, a single-element isolation method—such as an inflatable packer or a single retrievable packer or plug—is typically sufficient. However, when treating middle intervals, isolation of the lower and upper hydrocarbon-producing zones becomes necessary. In such cases, a tandem or dual element system is often required, combining an inflatable

packer above the treatment zone with a retrievable packer or plug set below to protect lower intervals.

- **Formation and Completion Pressure Limitations:** Reaching pressure limitations during injectivity tests—often caused by low formation injectivity—can pose a significant challenge in water shut-off operations. In such cases, a targeted pre-treatment, such as a small acid stimulation, may be necessary to enhance injectivity in the targeted zone. This approach improves fluid placement efficiency and facilitates the successful pumping of the water shut-off treatment.
- **Complexities in Rig-less Tool Deployment and Retrieval:** In completions with multiple restrictions, deploying and retrieving isolation tools becomes logistically demanding. Fishing operations may be necessary if tools become stuck, particularly in deviated wells.

Methodology

This section outlines the approach taken to plan, design, and execute coiled tubing water shut-off treatments. The methodology was developed to address the specific constraints of each completion type while leveraging recent advancements in non-polymer chemistry and downhole telemetry.

Candidate Selection and Pre-Job Planning

A comprehensive candidate selection was conducted for each candidate well. This included:

- Production Logging (PLT) including noise logs to verify existing interzonal barriers integrity: Tools such as PLTs, spinner surveys, and chemical tracer logs were used to identify the origin and intensity of water production. In some cases, historical well performance data was also leveraged.
- A detailed integration of geological and reservoir data was conducted, combining static models (e.g., porosity, permeability distributions) with dynamic reservoir parameters such as pore pressure and saturation. Completion records and mechanical properties were cross-referenced to identify intervals that are both accessible for intervention and viable for treatment, ensuring that targeted zones are treatable without compromising hydrocarbon-bearing sections.
- Diagnostics Planning: Real-time monitoring strategies were built into the planning phase to ensure downhole data could guide on-the-fly decision-making during execution.

Treatment Design

- Mechanical Isolation Tools: Selection of plugs, packers, or inflatable elements was based on available completion access and anchoring constraints.
- Chemical Design: Off & On-location laboratory testing was performed using samples and temperature simulations to select appropriate gel or non-polymer formulations. Temperature-activated systems were prioritized for improved control.
- Compatibility Testing: Fluid compatibility between formation water, treatment chemicals, and reservoir fluids was evaluated to minimize risks of precipitation or reaction.
- Volume Modeling: Reservoir geometry and annular capacity calculations were used to design the treatment volume with minimal excess.

BHA and Deployment Strategy

- BHA Design: Customized for each completion type to ensure optimal performance. Slim, pressure-resistant tools are used for ICD wells to navigate restricted spaces, while open hole completions demand compact, anchor-less assemblies to accommodate limited mechanical anchoring. Cased-hole perforated wells offer greater flexibility in BHA configurations due to better access and wellbore integrity.

- **Deployment Planning:** The coiled tubing string—considering diameter, metallurgy, and flexibility—and pump rates are carefully optimized to achieve precise fluid placement, minimize mechanical stress, and reduce the risk of equipment failure.
- **Contingency plans:** Comprehensive fishing tool inventories and well-specific backup plans are established in advance, particularly for wells with known retrieval challenges or complex geometries.
- **Operational Constraints:** Selection of BHA components also depends critically on their pressure and temperature ratings to ensure compatibility with well conditions and treatment requirements.

Real-Time Monitoring

- **Telemetry Integration:** Downhole sensors integrated into the BHA provided real-time measurements of pressure, temperature, and depth correlation using tools such as Casing Collar Locator (CCL) or gamma ray (GR) logs wherever feasible.
- **Data Utilization:** Continuous real-time data enabled confirmation of tool positioning, precise fluid placement, and timely initiation of treatment reactions. This immediate feedback allowed on-site adjustments when bottomhole conditions deviated from pre-job expectations, improving operational accuracy and success.
- **Temperature Monitoring:** Monitoring bottomhole temperature was especially critical for gel-based treatments that rely on temperature-triggered activation, ensuring the non-polymer gels formed at the intended depth and conditions.

Execution Framework and Operational Insights

This section outlines how the described methodology was implemented across different well completions, emphasizing best practices and the critical role of real-time data in guiding operational decisions. Rather than individual case histories, consolidated lessons learned from multiple jobs are presented.

A critical element across all water shut-off operations is the integration of downhole monitoring especially downhole temperature and injectivity testing. These practices enable real-time accurate adjustment of treatment formulations on location. This dynamic validation optimizes fluid placement, improves non-polymer activation, and ultimately enhances the effectiveness of the zonal isolation treatments.

Operations in ICD/Frac Port Wells

Water shut-off treatments in ICD completions require a conservative and precisely targeted approach due to the limited options for mechanical isolation. Slim, telemetry-enabled BHAs were deployed to continuously monitor tool position, bottomhole temperature, and downhole pressure, ensuring the ICD sleeves remained undisturbed throughout the operation. Gel-based fluids with delayed-setting properties were favored to allow controlled placement, while slow and carefully regulated pumping minimized the risk of over-displacement and treatment fluid migration. Furthermore, the pressure drop across the ICD nozzles during injection must be rigorously calculated and incorporated into the real-time pressure management and treatment placement strategy. Unlike production, where ICDs are designed to restrict flow into the wellbore, during injection, they restrict flowout of the wellbore, creating a significant pressure barrier. This injection pressure drop is a function of the fluid properties and the pump rate. Failure to account for this can lead to a critical miscalculation: the downhole pressure measured below the BHA may be within safe limits, but the resultant pressure in the casing annulus around the ICD could exceed the pressure rating of the completion jewelry, formation fracture pressure, or the integrity of the open-hole isolation packers. Therefore, pump rates must be optimized not only for placement accuracy but also to ensure the total pressure imposed on the completion components remains within their design limits, safeguarding the well's integrity throughout the

treatment. In some cases, a relatively high volume of shut-off material was necessary to achieve effective zonal isolation.

Best Practice: It is critical to identify the type, manufacturer, and pressure rating of the interzonal isolation barriers within the completion. Their integrity must be verified using noise logs or equivalent diagnostic tools before treatment. Since pressure ratings are typically based on the open hole diameter, applying an appropriate safety margin is essential when determining maximum injection pressures while considering the drop pressure across the completion flow restricting mechanisms. This precaution accounts for unknown conditions of the open hole diameter surrounding the barrier and helps prevent barrier failure during treatment.

Operations in Open Hole (Barefoot) Wells

Open hole completions carry inherent risks related to poor zonal isolation, largely due to the possible washouts and bad ovality along with the low success rate of open hole inflatable packers. To manage these challenges, precise placement in a section with good open hole inside diameter and ovality is critical. Injectivity tests are also needed during operations to monitor bottomhole pressure and assess the effectiveness of isolation tools in realtime. Pumping rates and sequences were continuously optimized based on live pressure data and fluid returns.

Best Practice: Prior to deploying open hole-compatible setting tools, borehole stability and diameter should be verified with detailed caliper logs. Additionally, using downhole gamma ray sensors helps confirm accurate CT BHA placement at the depth identified as having a suitable inner diameter, ensuring compatibility with BHA design limitations.

Operations in Screen Completions

Wells completed with a single screen are generally poor candidates for water shut-off treatments due to the absence of short zonal segmentation behind the screens to isolate selectively the water source. Conversely, wells with multiple screens separated by multiple interzonal isolation barriers represent suitable candidates for water shut-off interventions. As with other completion types, the BHA must be designed to navigate through all completion components, with the target zone providing a blank pipe or completion element suitable for anchoring the BHA.

Best Practice: Verifying the integrity of interzonal isolation barriers in wells with multiple screen intervals is essential to safeguard hydrocarbon-producing zones and enable selective water shut-off. Effective treatment design relies on close integration between detailed logging data interpretation and zonal targeting to overcome the inherent limitations of mechanical isolation in screen completions.

Operations in Perforated Cased Hole Wells

Perforated cased hole wells offer significant flexibility for mechanical isolation using tools such as bridge plugs, inflatable packers, or other zonal isolation devices. Pre-job production logging—including PLT and spinner surveys—enables precise identification and targeting of the perforated intervals responsible for water production. Real-time downhole telemetry is leveraged to confirm accurate tool placement and monitor fluid injection during treatment, enhancing operational control. Pre-treatment cleanout operations are crucial to ensure debris-free perforation surfaces, which improve sealing reliability and treatment effectiveness.

Best Practice: Verifying the integrity of the cement sheath behind the casing is crucial, especially when the target water-producing zone is close to hydrocarbon-producing intervals, to reduce the risk of crossflow behind the casing. Although continuously pumping through the annulus—when hydrocarbon zones are located above the target interval—can help mitigate crossflow, the risk remains. Additionally, this approach may dilute the water shut-off treatment, potentially reducing its effectiveness. However, it is generally preferable to taking no mitigating action.

Results & Discussion

The integrated results from multiple coiled tubing water shut-off operations confirm the effectiveness of tailored interventions in reducing water production while sustaining hydrocarbon output across various completion types. The combination of mechanical isolation and advanced non-polymer-based treatments, supported by real-time downhole monitoring, proved critical in achieving consistent zonal isolation and operational success.

Water Cut Reduction and Production Stabilization

- **Water cut reduction** post-treatment water cut reduction varies between **90% and 100%**, affected primarily by the treatment fluid volume.
- In many cases, oil production rates were maintained or even improved, benefiting from reduced water handling and surface treatment requirements.
- Successful operations minimized the risk of unintended production losses from non-target hydrocarbon zones.

Note: While the implemented water shut-off treatments have demonstrated significant reductions in water production, it is important to recognize that these treatments often provide temporary relief. As the reservoir continues to be subjected to waterflooding or natural water influx, water breakthrough can reoccur, necessitating additional interventions or treatment cycles to maintain production efficiency.

Key Observations

- **Pre-Treatment Logging:** The availability and quality of high-resolution PLT and temperature logs were decisive for success, especially in screen and open hole completions, where clear zonal boundaries are often absent.
- **Real-Time Data Utilization:** Downhole telemetry integration enabled dynamic operational adjustments such as modifying pump rates, fluid staging, or terminating treatments early in response to unexpected pressure or temperature variations.
- **Mechanical and Chemical Synergy:** Operations combining mechanical isolation devices (packers, plugs) with non-polymer-based chemical treatments delivered improved zonal sealing, enhanced fluid placement accuracy, and long-term stability of the shut-off.
- **Cleanout Operations:** Pre-treatment cleanouts proved essential to remove debris, scale, and other obstructions, thereby facilitating proper mechanical tool seating and improving treatment effectiveness as well as post-treatment cleanout to ensure wellbore accessibility for future interventions.

Economic Impact

Rig-less coiled tubing interventions demonstrated consistent advantages over traditional rig-based methods, including:

- Reduced non-productive time due to faster mobilization and job execution.
- Lower intervention costs per barrel of incremental oil produced or water deferred.
- Enhanced operational flexibility enabling rapid response to reservoir conditions without the need for rig mobilization.

Best Practices and Lessons Learned

Based on cumulative operational experience, the following best practices are recommended to optimize future coiled tubing water shut-off campaigns:

Operational Checklist

- Verify well integrity thoroughly before intervention.
- Utilize high-resolution production logging and PLT for precise water zone identification.
- Confirm chemical compatibility of treatment fluids with formation water and reservoir conditions.
- Customize BHA design to match specific completion architectures and downhole constraints.
- Conduct comprehensive pre-job cleanout and calibration runs to ensure debris-free and stable wellbore conditions.
- Continuously monitor downhole temperature, pressure, and depth during treatment to confirm accurate placement and gel activation.
- Avoid over-displacement, especially in open hole and screen completions, to prevent fluid migration and treatment dilution.

Key Success Factors

- Achieve high-quality zonal isolation by accurately identifying and sealing water-producing intervals.
- Perform on-site chemistry validation to ensure non-polymers gels activate as intended under actual downhole temperature and pressure.
- Leverage real-time telemetry data to dynamically respond to operational variations and optimize fluid placement.
- Implement post-treatment surveillance through follow-up logging or production monitoring to assess treatment success and inform repeat interventions if needed.

Conclusion

Rig-less coiled tubing water shut-off operations represent a highly effective, cost-efficient, and flexible solution for managing increasing water production in mature reservoirs. By tailoring treatment designs and BHAs to the specific completion types—whether perforated cased hole, ICD-equipped, screens, or open hole wells—and leveraging real-time downhole monitoring combined with on-location chemical optimization, operators can significantly enhance shut-off success. Furthermore, the study underscores the critical importance of understanding completion-specific injection mechanics, such as managing the pressure drop across inflow control devices to safeguard well integrity during treatment placement.

This integrated approach reduces surface water handling burdens and operational costs, while enabling faster and safer interventions without rig mobilization. Ultimately, these best practices, which encompass both advanced chemical solutions and detailed pressure management strategies for complex completions, support sustained oil production, improved reservoir management, and enhanced field economics.

It should be noted that water shut-off treatments typically offer temporary mitigation of water production. Due to the ongoing dynamics of reservoir flooding, water breakthrough may recur after a period, requiring repeated interventions. Nonetheless, the techniques presented provide a cost-effective and flexible approach to managing water production and sustaining hydrocarbon output in mature fields.

Acknowledgment

The Authors would like to thank all TAQA's Coiled Tubing Operation team in making such an achievement possible with their dedication and resilience.

Nomenclature

CT = Coiled Tubing
BHA = Bottomhole Assembly
ICD = Inflow Control Device
WSO = Water Shut-off
CCL = casing collar locator

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